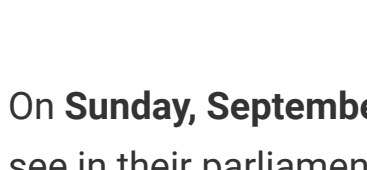


Data vis do's & don'ts
10 min
August 27th, 2021

How to visualize polls and results of the German election with Datawrapper

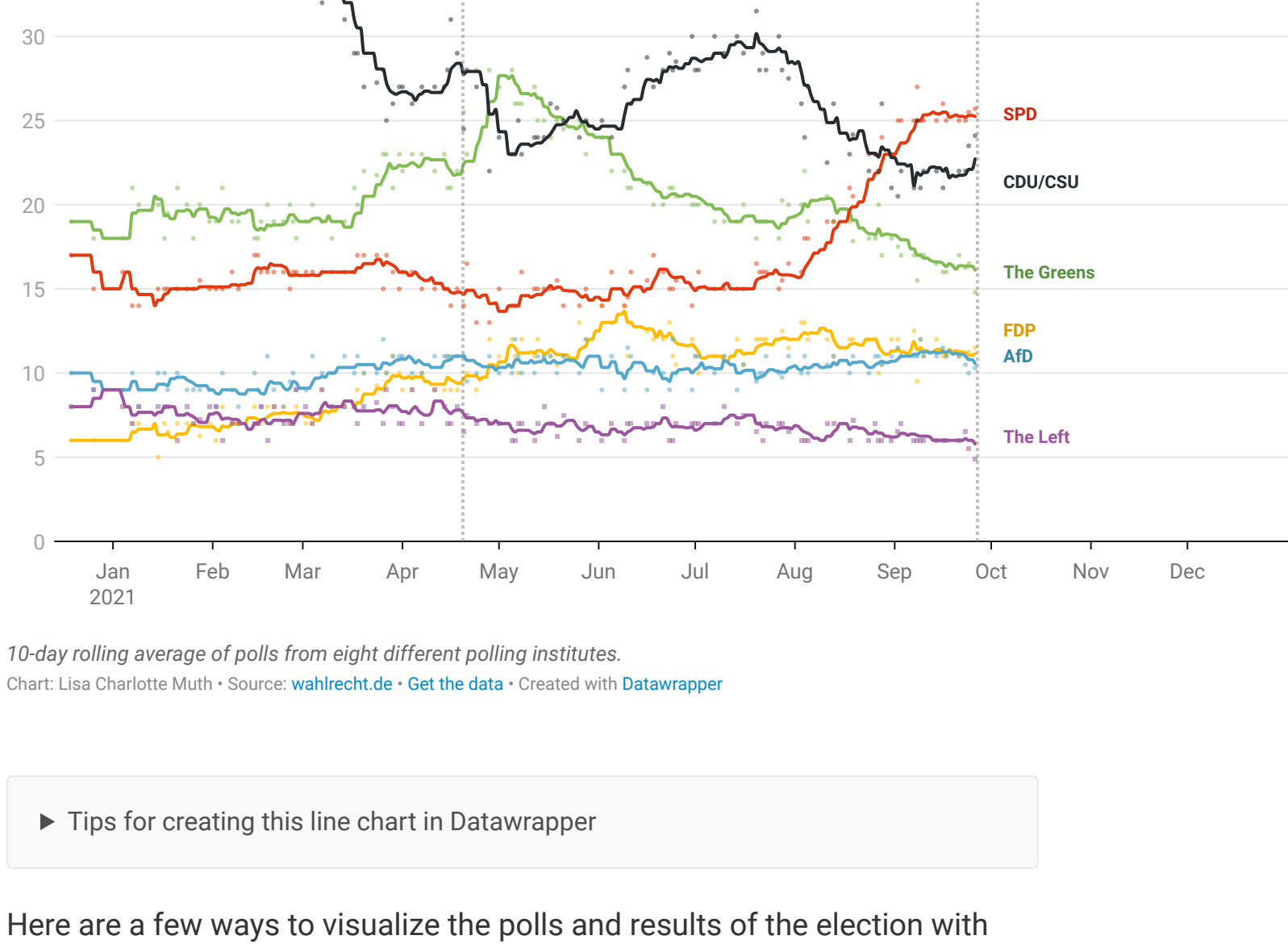


Lisa Charlotte Muth

On **Sunday, September 26, 2021**, Germans will vote for who they want to see in their parliament for the next four years, and, indirectly, for Angela Merkel's successors.

(To learn how the German election system works in detail, visit my [casual, visual explanation from 2017](#), the [shorter, also visual Bloomberg explanation from 2017](#), or [this explainer from the Economist from a few weeks ago](#).)

The last few months have been interesting. Polls show lots of undecided people: Almost 40% of German voters said they'd vote for the CDU in July last year, but now that's down to less than 25%.



10-day rolling average of polls from eight different polling institutes.
Chart: Lisa Charlotte Muth • Source: [wahlrecht.de](#) • [Get the data](#) • Created with [Datawrapper](#)

► Tips for creating this line chart in Datawrapper

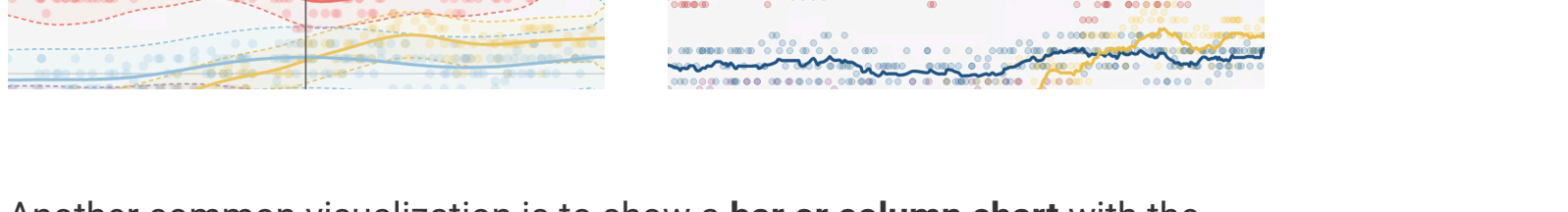
Here are a few ways to visualize the polls and results of the election with Datawrapper.

Similar to the [U.S. elections last year](#), we won't offer any **live-updating charts on election night**. However, the friendly people from [wahlrecht.de](#) gather polls from eight different institutions, and we'll use them in most of the following charts. They will check for new polls multiple times each day. **Feel free to re-use all of the following visualizations**, no matter if they come with live-updating [wahlrecht.de](#) data or the old 2017 data: Hover over the visualization of your choice, click on "Edit this chart," change the data if necessary, and then publish and embed it on your own website.

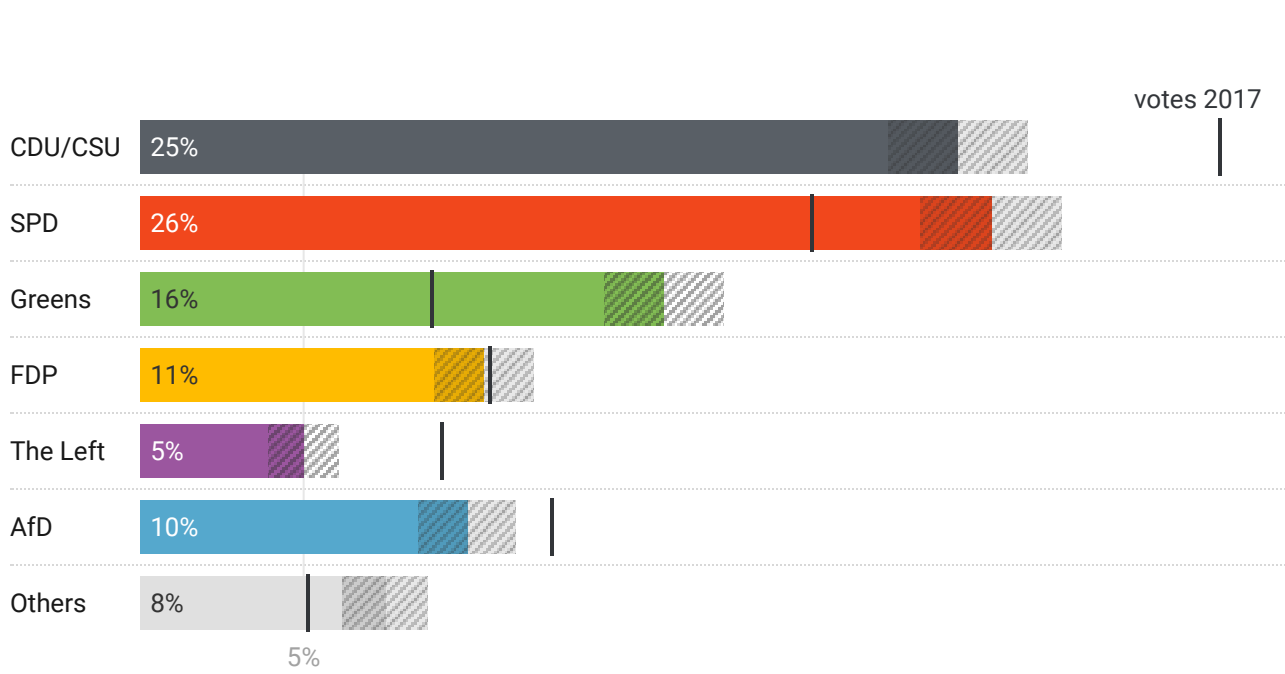
How many votes will each party get?

There are a few ways to show the latest polls. The one that leads most articles (e.g., from [ZEIT, NZZ, Tagesspiegel, The Guardian](#), or [Politico](#)) is the **line chart that shows polls over time** – sometimes just for the year 2021, sometimes since 2017.

Newsrooms can choose between different ways of averaging polls (e.g., [moving average](#) or [Loess](#)) and different levels of smoothing – the more polls are averaged, the less ragged the lines become. Compare the smooth [Economist](#) lines with the ragged [Guardian](#) lines:



Another common visualization is to show a **bar or column chart** with the latest poll (ZDF), an average of the latest polls (SPIEGEL), or a forecast (The Economist). Showing confidence intervals when choosing this view helps to communicate the data's uncertainty and the fact that these are not the final results.



Latest poll about the 2021 German federal election of 2021. If multiple polls are released per day, they're averaged. The polling institute is 95% certain that German public opinion is somewhere in the shaded area.
Source: [wahlrecht.de](#) • [Get the data](#) • Created with [Datawrapper](#)

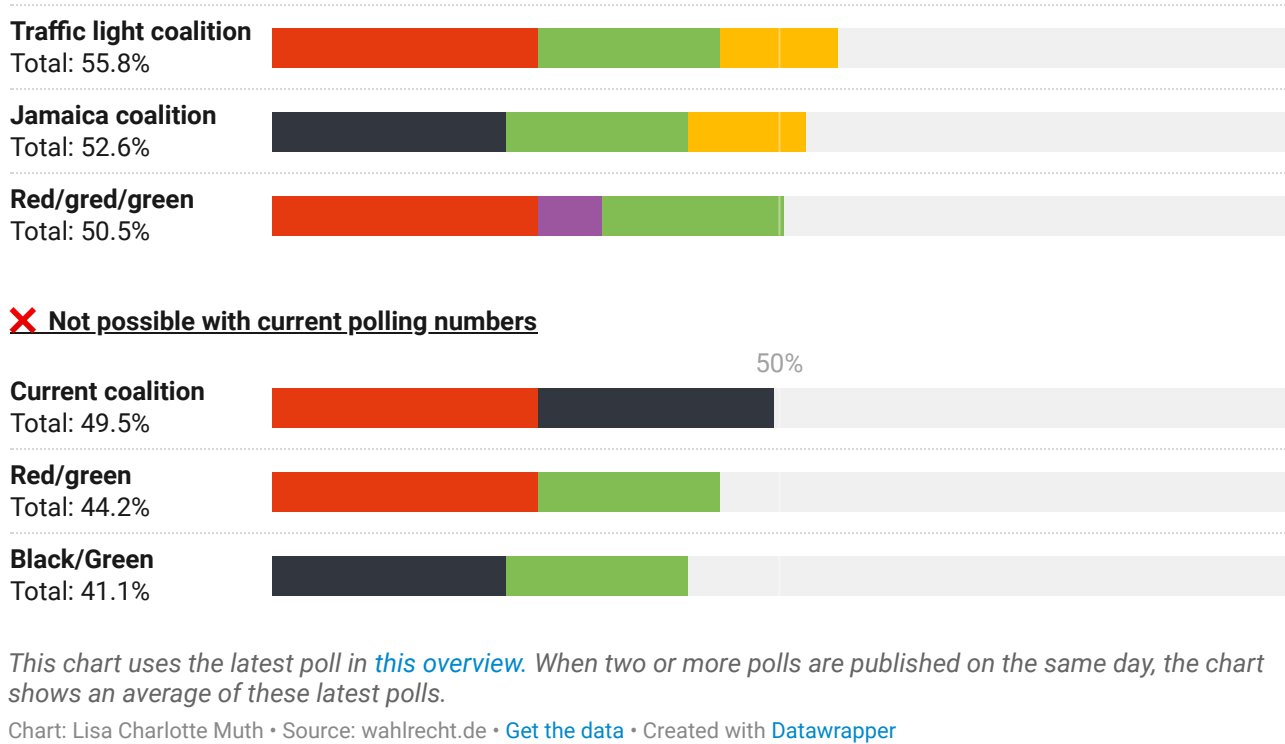
► Tips for creating confidence intervals in Datawrapper bar charts

Which coalitions will be possible?

To govern, parties need a majority in the German parliament. There's no party that can convince 50% of German voters to vote for them these days, so the only way to achieve this is to play buddies with other parties and form a **coalition**. This chart visualizes which coalitions are possible, based on the latest poll. (I didn't come up with the weird coalition names; they're commonly used!)

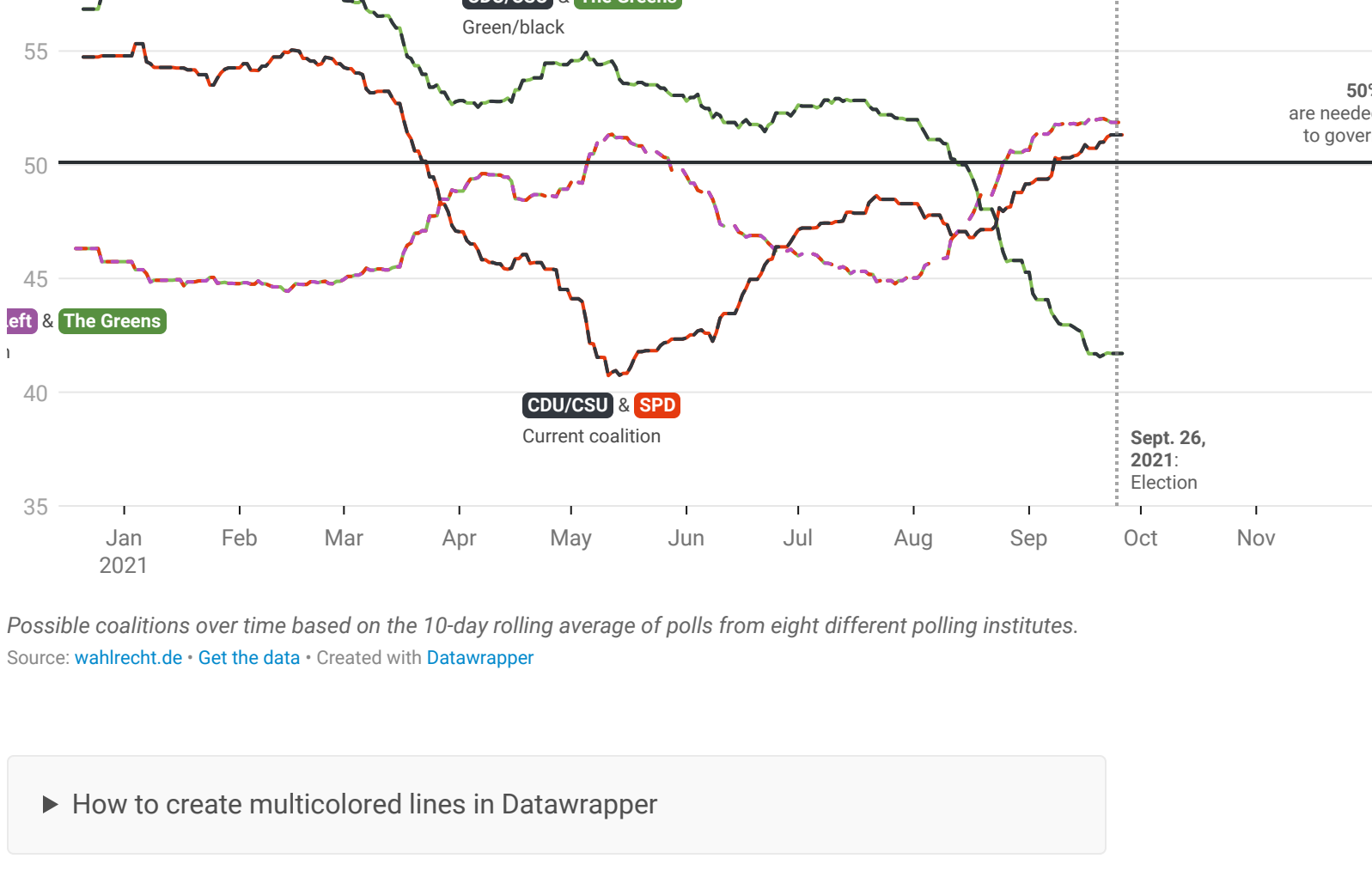
50% are needed to govern

If a party gets less than 50% of the votes, it needs to join forces with one or more other parties to become the governing coalition. These are the mathematically possible and impossible coalitions according to the latest polling numbers:



This chart uses the latest poll in [this overview](#). When two or more polls are published on the same day, the chart shows an average of these latest polls.
Chart: Lisa Charlotte Muth • Source: [wahlrecht.de](#) • [Get the data](#) • Created with [Datawrapper](#)

Like polls, we can also show the possible coalitions over time. Here's my experimental attempt, inspired by [SPIEGEL](#):



Possible coalitions over time based on the 10-day rolling average of polls from eight different polling institutes.
Source: [wahlrecht.de](#) • [Get the data](#) • Created with [Datawrapper](#)

► How to create multicolored lines in Datawrapper

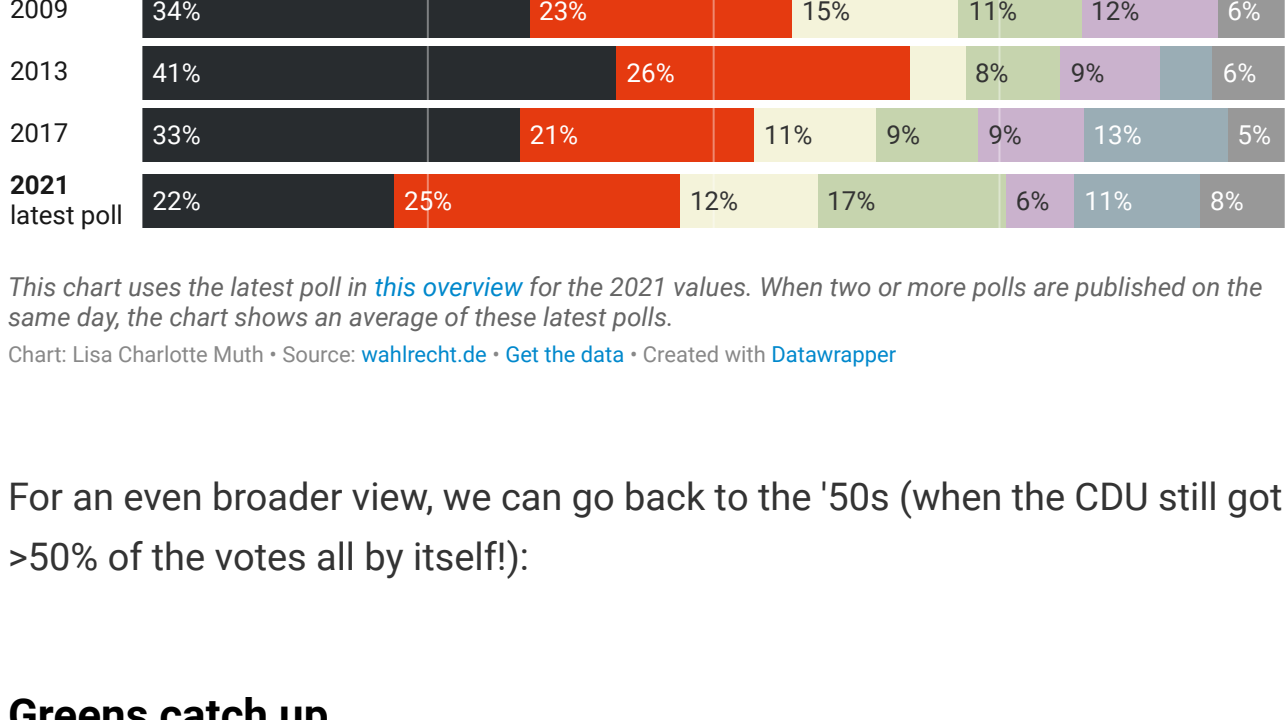
Next, let's zoom a bit out and look at recent decades:

How do these polls look in historical context?

We can compare the latest poll(s) with the results from the past few elections to recognize trends. Here's a chart showing the rise of traditionally small parties – led by the Greens – and the loss of trust in the two big parties, CDU/CSU and SPD:

Smaller big parties

German federal election results since 1990. 2021 might be the first election since 1990 where CDU/CSU and SPD don't get >50% of the votes in total.

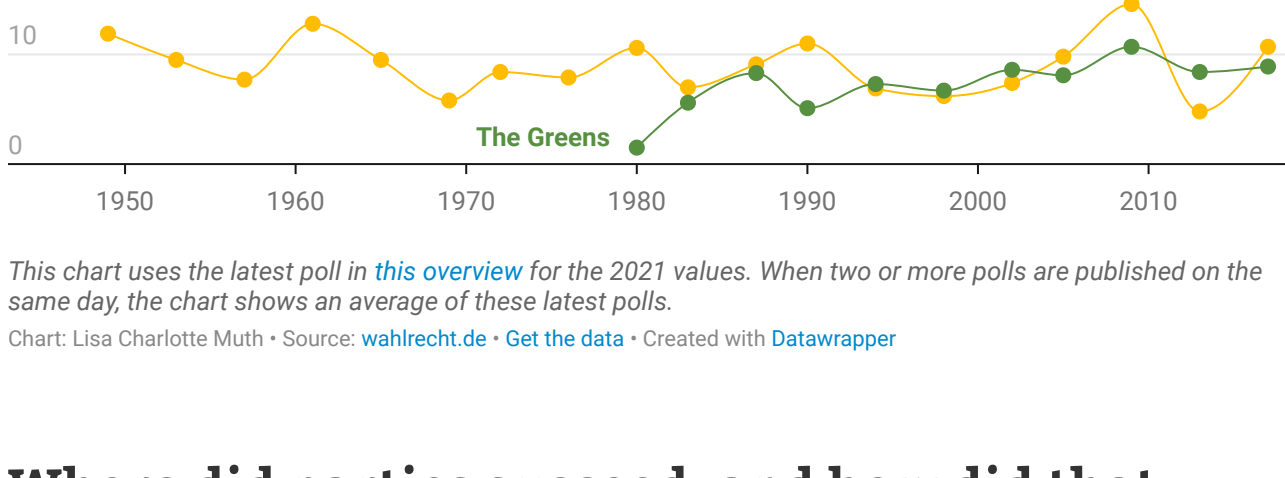


This chart uses the latest poll in [this overview](#) for the 2021 values. When two or more polls are published on the same day, the chart shows an average of these latest polls.
Chart: Lisa Charlotte Muth • Source: [wahlrecht.de](#) • [Get the data](#) • Created with [Datawrapper](#)

For an even broader view, we can go back to the '50s (when the CDU still got >50% of the votes all by itself):

Greens catch up

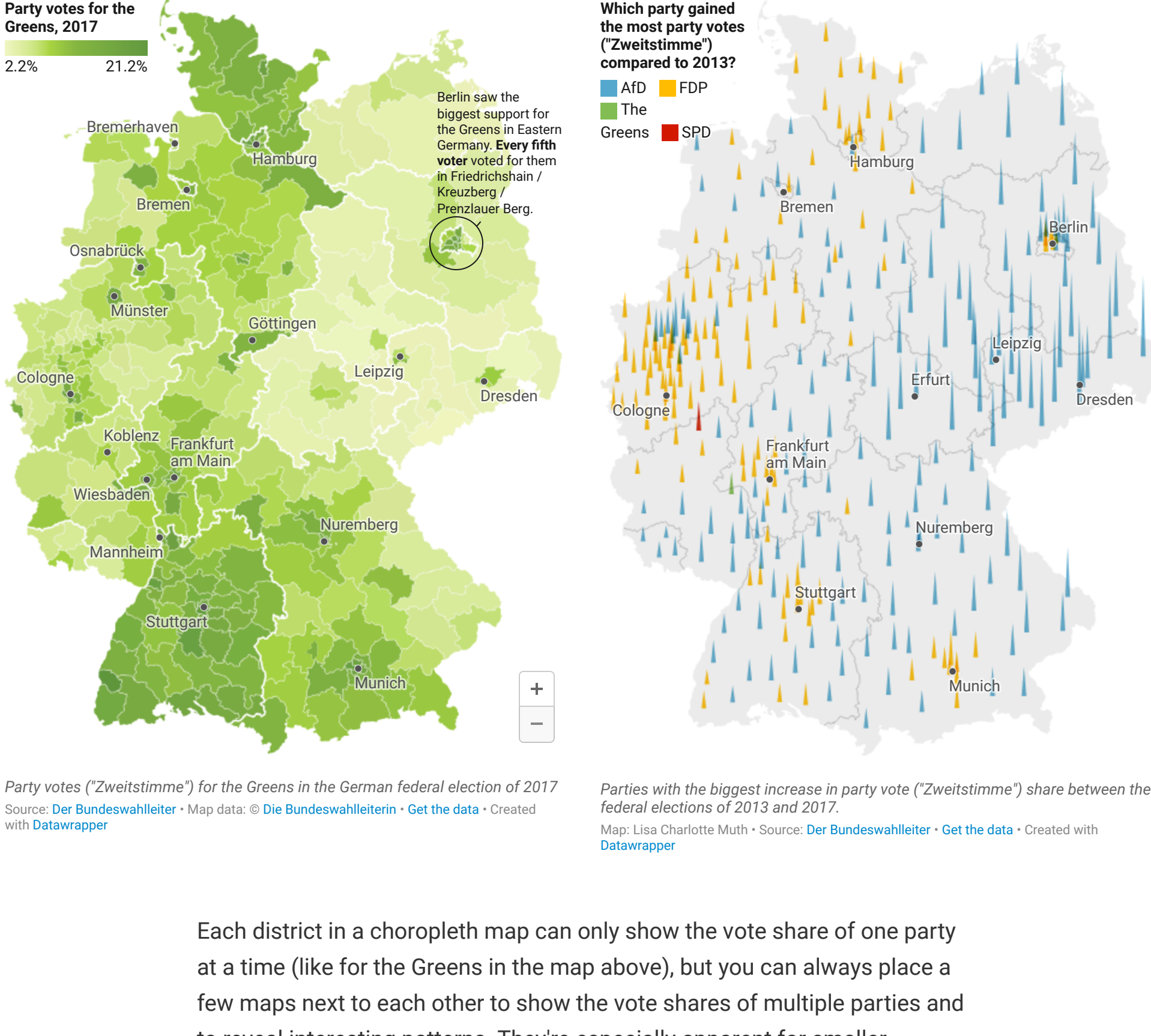
Election results since 1950 for CDU/CSU, SPD, FDP, and Greens. 2021 numbers are not the final results, but a poll from late August 2021.



This chart uses the latest poll in [this overview](#) for the 2021 values. When two or more polls are published on the same day, the chart shows an average of these latest polls.
Chart: Lisa Charlotte Muth • Source: [wahlrecht.de](#) • [Get the data](#) • Created with [Datawrapper](#)

Where did parties change, and how did that change?

When it comes to visualizing the results, we can show not just the normal bar/column chart with the party vote shares, but also their regional distribution. The following map shows the vote shares of the Green party in 2017. The second map shows which party got the most new voters in each election district compared with the 2013 election:



Party votes ("Zweitstimme") for the Greens in the German federal election of 2017
Source: [Der Bundeswahlleiter](#) • Map data: © [Die Bundeswahlleiter](#) • [Get the data](#) • Created with [Datawrapper](#)

Parties with the biggest increase in party vote ("Zweitstimme") share between the federal elections of 2013 and 2017
Map: Lisa Charlotte Muth • Source: [Der Bundeswahlleiter](#) • [Get the data](#) • Created with [Datawrapper](#)

Each district in a choropleth map can only show the vote share of one party at a time (like for the Greens in the map above), but you can always place a few maps next to each other to show the vote shares of multiple parties and to reveal interesting patterns. They're especially apparent for smaller parties, as [The Berliner Morgenpost](#) showed in 2017.

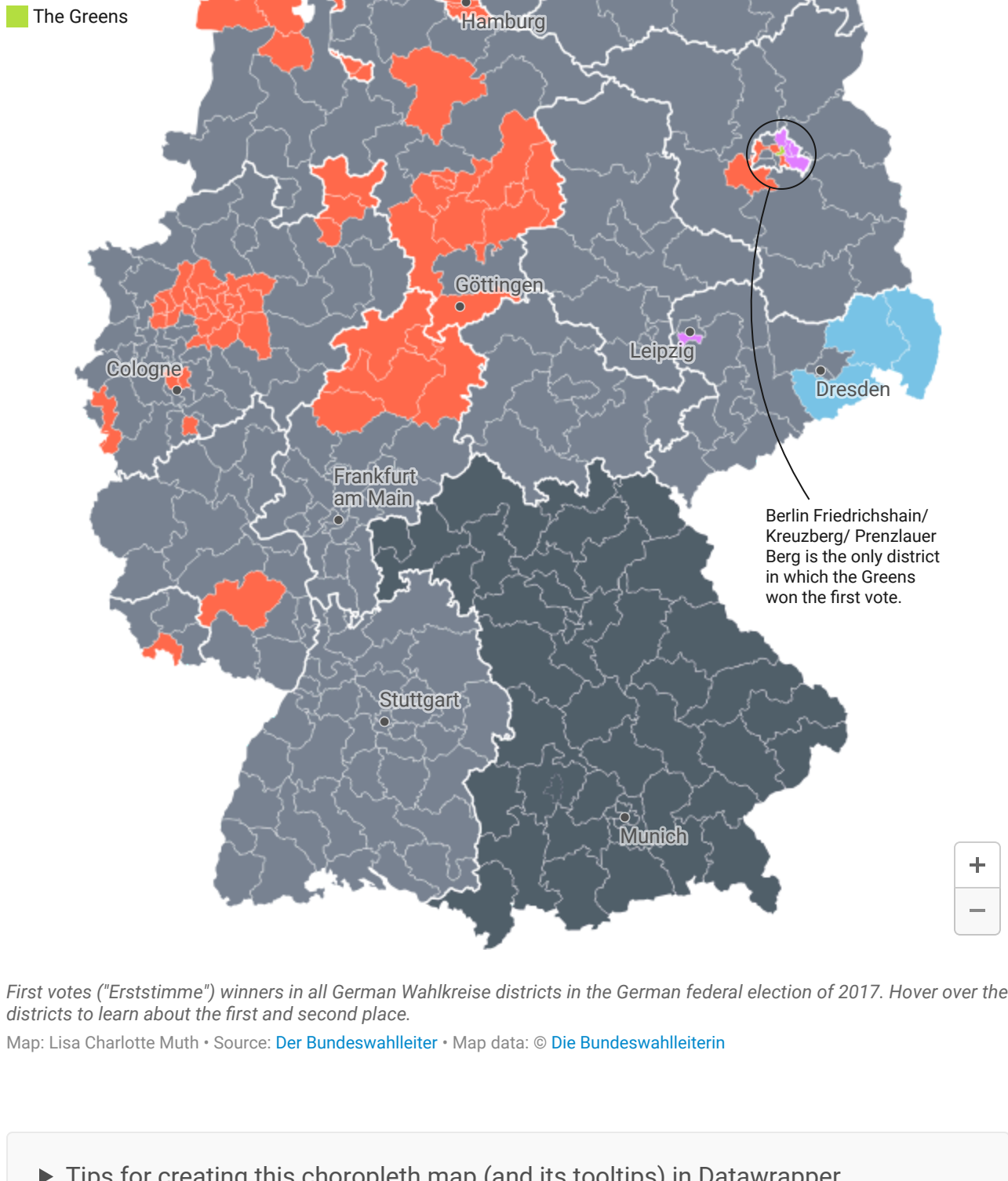
You can also compare two parties in one choropleth map. I wrote about it in-depth a few years ago in [this Weekly Chart](#) that visualizes votes of both the Left and the AfD on a map of Berlin's Wahlbezirke districts.

Who won the "first vote"?

Last and least: The "first vote" ("Erststimme"). It sounds important – but while the party vote ("Zweitstimme") decides the crucial share of parties in the parliament, the first vote "only" decides which party representative in your election district will be part of that parliament.

The first vote works as a winner-take-all system: The person with the most votes will get in, the others won't.

Most newsrooms show the regional distributions of the first vote with a **choropleth map**. Here's one with the data for 2017:



First votes ("Erststimme") winners in all German Wahlbezirke districts in the German federal election of 2017. Hover over the districts to learn about the first and second place.
Map: Lisa Charlotte Muth • Source: [Der Bundeswahlleiter](#) • Map data: © [Die Bundeswahlleiter](#)

► Tips for creating this choropleth map (and its tooltips) in Datawrapper

One of the more interesting aspects to explore is **the margin for the unlucky candidate who got second place**. I decided to put that information into the tooltip. The vote shares of the winner and 2nd-place parties are fairly close in some areas – try to hover over the blue-colored [Görlitz](#), the easternmost election district in Germany, where the AfD candidate won with only 1,578 more votes than the CDU candidate.

We hope that these visualizations provide a good starting point for your election visualizations. To use any of the ones above, hover over them and click on "Edit this chart." This will lead you right to step 3: Visualize in the Datawrapper app. From there, you can edit the chart (by changing the data source, title, labels, colors, etc.) or just publish it directly to get an embed code for your site. Any questions? Let us know at support@datawrapper.de.

Lisa Charlotte Muth (she/her: @lisacmuth, @lisacmuth@vis.social) is Datawrapper's head of communications. She writes about best practices in data visualization and thinks of new ways to excite you about charts and maps. Lisa lives in Berlin.

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